

Zinc-Diethylaluminium Chloride Induced Coupling Reaction of Dibromofluoroacetate
with Carbonyl Compounds. New Efficient and Selective Synthesis of
 α -Bromo- α -fluoro- β -hydroxy and α -Fluoro- β,β' -dihydroxy Esters¹⁾

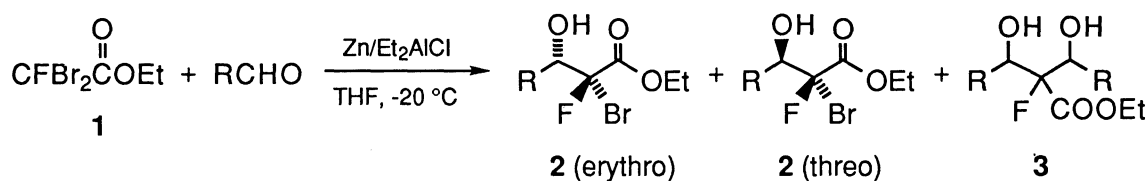
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Treatment of ethyl dibromofluoroacetate with aldehydes or ketone in the presence of zinc and diethylaluminium chloride at -20 °C gave rise to the corresponding α -bromo- α -fluoro- β -hydroxy alkanolic acid ethyl esters in good yields, while the use of two equivalents each of aldehyde, zinc, and diethylaluminium chloride in the reaction resulted in a double coupling reaction affording the 1:2 adducts, α -fluoro- β,β' -dihydroxy esters in high yields.

The Reformatsky reaction of α -halo esters with carbonyl compounds is one of the most important carbon-carbon bond forming reactions and has widely been applied for the synthesis of β -hydroxy alkanolic acid esters, which serve as a useful building block in constructing a variety of naturally occurring substances. Although a number of methods for the reactions of this type have been reported in the literature,²⁾ little is known for the reaction between a di- or trihaloacetate and a carbonyl compound leading respectively to α -halo- or α,α -dihalo- β -hydroxy esters.³⁾ To our best knowledge,⁴⁾ there exist only two contributions, made by Castro *et al.*^{3a)} and Fürstner,^{3b)} that pertain to a classical or modified procedure for effecting successfully the Reformatsky reaction of dichloro-, dibromo-, or trichloroacetate.

This communication describes a new efficient and convenient method for the reductive coupling reaction of dibromofluoroacetate (**1**) with carbonyl compounds induced by zinc and diethylaluminium chloride,⁵⁾ which provides us with the first promising approach for synthesizing selectively α -monofluorinated α -bromo- β -hydroxy (**2**) or β,β' -dihydroxy esters (**3**) in good yields.



When ethyl dibromofluoroacetate (**1**) was allowed to react with butyraldehyde in the presence of zinc and diethylaluminium chloride in tetrahydrofuran (THF) at -20 °C for 1 h, the corresponding 1:1 adduct, α -bromo- α -fluoro- β -hydroxy hexanoic acid ethyl ester (**2b**),⁷⁾ was obtained as a mixture of erythro and threo isomers^{8,9)}

Table 1. Synthesis of Hydroxy Esters **2** via Coupling Reaction of **1** with Carbonyl Compounds

Entry	Carbonyl compd	Yield/(% ^a) of 2		Isomer ratio of 2 erythro : threo ^b	Yield/(% ^a) of 3	
1	CH ₃ CH ₂ CHO	2a	63	66 : 34	3a	14
2	CH ₃ CH ₂ CH ₂ CHO	2b	69	67 : 33	3b	13
3	(CH ₃) ₂ CHCHO	2c	70	67 : 33	3c	2
4	<i>c</i> -C ₆ H ₁₁ CHO	2d	67	63 : 37	3d	14
5	CH ₃ CH=CHCHO	2e	68	55 : 45	3e	19
6	C ₆ H ₅ CHO	2f	71	59 : 41	3f	8
7	<i>p</i> -CH ₃ C ₆ H ₄ CHO	2g	77	51 : 49	3g	3
8	<i>p</i> -CH ₃ OC ₆ H ₄ CHO	2h	74	52 : 48	3h	2
9	<i>p</i> -ClC ₆ H ₄ CHO	2i	66	63 : 37	3i	15
10	2-furaldehyde	2j	74	59 : 41	3j	10
11	CH ₃ CH ₂ COCH ₂ CH ₃	2k	49	—	4k	(15) ^c

a) Yields refer to analytically pure products isolated by column chromatography. b) The isomer ratios were determined by ¹⁹F NMR. c) The figure in parentheses stands for the yield of ethyl (Z)-3-ethyl-2-fluoro-2-pentenoate (**4k**).

in 69% yield, together with a small amount of the 1:2 adduct, α -fluoro- β,β' -dihydroxy ester **3b**⁷⁾ (13%). Diethylaluminium chloride was essential for the reaction to proceed cleanly. The absence of this Lewis acid led to the formation of substantial amounts of by-products including diethyl difluorofumarate and difluoromaleate,⁷⁾ and the yield of the adduct **2b** became low (36%). Other Lewis acids, such as triethylaluminium and boron trifluoride diethyl etherate, were much less satisfactory than diethylaluminium chloride. The reaction carried out below -20 °C (at -30 or -50 °C) took place very sluggishly, whereas the reaction at temperatures higher than -20 °C resulted in the formation of **3b** (30%), while **2b** was formed in 32% yield. THF was a favored solvent of the reaction. The use of diethyl ether or benzene caused no reaction, and the starting acetate **1** was recovered unchanged.

The following procedure is representative for the coupling reaction of **1** with carbonyl compounds.¹¹⁾ Diethylaluminium chloride (1.1 equiv., a 1.0 mol dm⁻³ hexane solution) and a carbonyl compound (1.1 equiv.) were sequentially introduced into a stirred suspension of zinc dust (1.2 equiv.) in THF at -20 °C under argon. To this mixture was added dropwise a solution of **1** in THF at such a rate that the reaction temperature does not rise above -15 °C. After stirring for 1 h at -20 °C, the reaction mixture was hydrolyzed with cold 10% HCl and saturated NH₄Cl solutions and then was extracted with diethyl ether. The extracts were dried over Na₂SO₄ and concentrated *in vacuo* to give an oil, which was chromatographed on silica gel by successive use of hexane-EtOAc (2 : 1) and EtOAc to obtain β -hydroxy ester **2**⁷⁾ and β,β' -dihydroxy ester **3**.⁷⁾ The results of these reactions are summarized in Table 1.

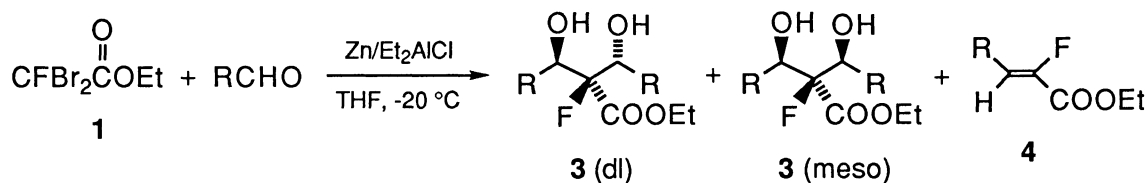
Various aliphatic (Entries 1-4) and aromatic aldehydes (Entries 6-10) including α,β -unsaturated aldehyde (Entry 5) could participate nicely in the reaction to afford the expected adducts, α -bromo- α -fluoro- β -hydroxy esters **2a-j**, in good yields as a diastereomeric mixture. Their erythro to threo ratios range from 51/49 to 67/33.

Table 2. Synthesis of Dihydroxy Esters **3** through Coupling Reaction of **1** with Aldehydes

Entry	Aldehyde	Time/h	Yield/(% ^a) of 3		Isomer ratio of 3 dl : meso ^b)	Yield/(% ^a) of 4	
12	CH ₃ CH ₂ CHO	3	3a	73	63 : 37	4a	14
13	CH ₃ CH ₂ CH ₂ CHO	3	3b	79	64 : 36	4b	12
14	(CH ₃) ₂ CHCHO	3	3c	77	66 : 34	4c	10
15	CH ₃ CH=CHCHO	2	3e	76	56 : 44	4e	15
16	C ₆ H ₅ CHO	2	3f	79	64 : 36	4f	20
17	<i>p</i> -CH ₃ C ₆ H ₄ CHO	2	3g	74	61 : 39	4g	17
18	<i>p</i> -CH ₃ OC ₆ H ₄ CHO	2	3h	73	66 : 34	4h	14
19	<i>p</i> -ClC ₆ H ₄ CHO	2	3i	70	57 : 43	4i	20

a) Yields are of pure products isolated by column chromatography. b) The isomer ratios were measured by ¹⁹F NMR.

The coupling reaction with a ketone such as 3-pentanone (Entry 11) also occurred to give the corresponding adduct **2k** in moderate yield, along with 15% yield of ethyl (Z)-3-ethyl-2-fluoro-2-pentenoate (**4k**),⁷⁾ no dihydroxy ester **3k** being produced at all. It may be noted that, while the present reaction was accompanied by the formation of small portions (2-19%) of dihydroxy esters **3a-j** or α -fluoro- α,β -unsaturated ester **4k**, these side products were easily removed by column chromatography.



It was found, moreover, that simple change in a molar ratio of the reagents made it possible to produce dihydroxy esters **3** predominantly. Thus, on treating **1** with aldehydes (2.1 equiv.) in the presence of zinc (2.1 equiv.) and diethylaluminium chloride (2.1 equiv.) at -20 °C for 2 or 3 h, the corresponding 1:2 adducts **3**⁷⁾ were given as a mixture of two diastereoisomers (dl : meso = 56/44 to 66/34) in high yields and (Z)- α -fluoro- α,β -unsaturated esters **4**⁷⁾ were formed in less than 20% yield as a sole by-product (Table 2).

Interestingly, the monitoring of the reaction between **1** and zinc (1.2 equiv.) without carbonyl compound by ¹⁹F NMR indicated that the resonance due to **1** at δ 12.3 (downfield from external trifluoroacetic acid (TFA)) disappeared and a new resonance appeared at δ -61.0. Hydrolysis of this mixture with deuterium oxide (D₂O) offered ethyl deuteriobromofluoroacetate.^{7,12)} Similarly, treatment of **1** with 2 equiv. of zinc followed by quenching with D₂O gave ethyl dideuteriofluoroacetate.^{7,12)} These facts strongly suggest that the former and the latter reaction would involve an intermediary monometallated and dimetallated^{5a)} species, respectively, though their exact structures are unclear at the present stage.

In summary, the Reformatsky-type aldol reaction of dibromofluoroacetate (**1**) with various carbonyl compounds has been shown to proceed efficiently under the influence of zinc and diethylaluminium chloride.

The present reaction will constitute an effective and practical synthetic method for α -fluorinated β -hydroxy **2** or β,β' -dihydroxy esters **3**, which are hard to obtain by other methods.

References

- 1) Presented in part at the 14th International Symposium on Fluorine Chemistry, Yokohama, Japan, August 1994, Abstr. No. 1P-49.
- 2) a) M. W. Rathke, *Org. React.*, **22**, 423 (1975); b) A. Fürstner, *Synthesis*, **1989**, 571.
- 3) a) B. Castro, J. Villieras, and N. Ferracutti, *Bull. Soc. Chim. Fr.*, **1969**, 3521; b) A. Fürstner, *J. Organomet. Chem.*, **336**, C33 (1987); c) for the reaction of α,α -dibromo ketones, A. Takahashi and M. Shibasaki, *J. Org. Chem.*, **53**, 1227 (1988).
- 4) Fluorine-containing haloacetates are shown to undergo the Reformatsky reaction giving α -fluorinated β -hydroxy alkanoates. For the reactions of bromofluoroacetates, see: E. T. McBee, O. R. Pierce, and D. L. Christman, *J. Am. Chem. Soc.*, **77**, 1581 (1955); S. Brandänge, O. Dahlman, and L. Mörch, *ibid.*, **103**, 4452 (1981). For the reactions of chloro-, bromo-, or iododifluoroacetates and of related carbonyl compounds, see: E. A. Hallinan and J. Fried, *Tetrahedron Lett.*, **25**, 2301 (1984); T. Yokozawa, N. Ishikawa, and T. Nakai, *Chem. Lett.*, **1987**, 1971; R. W. Lang and B. Schaub, *Tetrahedron Lett.*, **29**, 2943 (1988); O. Kitagawa, T. Taguchi, and Y. Kobayashi, *ibid.*, **29**, 1803 (1988); M. Kuroboshi and T. Ishihara, *Bull. Chem. Soc. Jpn.*, **63**, 428 (1990); T. Tsukamoto and T. Kitazume, *J. Chem. Soc., Chem. Commun.*, **1992**, 540; T. T. Curran, *J. Org. Chem.*, **58**, 6360 (1993), and references cited therein.
- 5) It has been reported that the coupling reaction of polyhaloacetates with carbonyl compounds in the presence of zinc-diethylaluminium chloride leads to the exclusive formation of α,β -unsaturated esters.⁶⁾ See: a) K. Takai, Y. Hotta, K. Oshima, and H. Nozaki, *Bull. Chem. Soc. Jpn.*, **53**, 1698 (1980); b) A. R. Daniewski and W. Wojciechowska, *J. Org. Chem.*, **47**, 2993 (1982), and see also Ref. 2b.
- 6) For the related reactions of polyfluorohalo esters, see: T. Ishihara and M. Kuroboshi, *Chem. Lett.*, **1987**, 1145; T. Allmendinger and R. W. Lang, *Tetrahedron Lett.*, **32**, 339 (1991).
- 7) Spectroscopic and analytical data of the products were in good accord with the assigned structures.
- 8) The relative stereochemical nomenclature proposed by Noyori *et al.* is applied in this work. See: R. Noyori, I. Ishida, and J. Sakata, *J. Am. Chem. Soc.*, **105**, 1598 (1983).
- 9) The configurational assignment of the products **2** or **3** was made on the basis of the relative magnitudes of vicinal hydrogen-fluorine couplings.¹⁰⁾ For instance, ¹H NMR (CDCl₃) of the acetonide derived from **2f** (erythro : threo = 59 : 41): δ 1.46 (br s, 6H x 2), 3.4-4.4 (m, 2H x 2), 4.77 (d, J = 8.0 Hz, 1H) for the erythro isomer, 4.91 (d, J = 24.0 Hz, 1H) for the threo isomer, 7.1-7.4 (br s, 5H x 2).
- 10) T. Yamazaki, T. Yamamoto, and T. Kitazume, *J. Org. Chem.*, **54**, 83 (1989); M. Kuroboshi and T. Ishihara, *Bull. Chem. Soc. Jpn.*, **63**, 1191 (1990); T. Ishihara, M. Kuroboshi, K. Yamaguchi, and Y. Okada, *J. Org. Chem.*, **55**, 3107 (1990).
- 11) The reaction procedure could be applied to a large-scale (10 mmol) preparation. Zinc dust was washed sequentially with 10% HCl, methanol, and diethyl ether, and was dried under vacuum.
- 12) ¹⁹F NMR data (from external TFA): CDFBrCOOEt (47%) δ -73.0 (t, J = 6.8 Hz); CD₂FCOOEt (57%) δ -154.0 (quint, J = 7.5 Hz). For comparison, CHFBrCOOEt δ -72.0 (d, J = 46.9 Hz); CH₂FCOOEt δ -152.5 (t, J = 47.1 Hz).

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